

Control of Large-Scale Irrigation Networks

With sensors, actuators, information processing and communications, open-water channels regulated by gates can be modeled and managed by optimal closed-loop control.

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ABSTRACT | Irrigation networks of open-water channels are used throughout the world to support agricultural activity. By and large, these networks are managed in open loop. To achieve closed-loop water distribution management, it is necessary to augment these civil engineering systems with an appropriate information infrastructure—sensors, actuators, information processing, and communication resources. Recent pilot projects in Australia demonstrate the significant potential of closed-loop management, which can yield a significant improvement in the quality of service, while achieving improved water distribution efficiency. This paper focuses on the modelling and closed-loop control of open-water channels from the perspective of large-scale irrigation network management. Several feedback information structures are discussed and the key design tradeoffs identified.

KEYWORDS | Distributed control; feedback controller design; irrigation networks; open-water channels

I. INTRODUCTION

On a global scale, irrigation accounts for approximately 70% of all fresh water usage [1]. The distribution of fresh water for irrigation is often achieved via an extensive civil infrastructure of reservoirs (both natural and man made) and open-water channels, which serve the farms, as illustrated in Fig. 1. In large-scale networks, there is

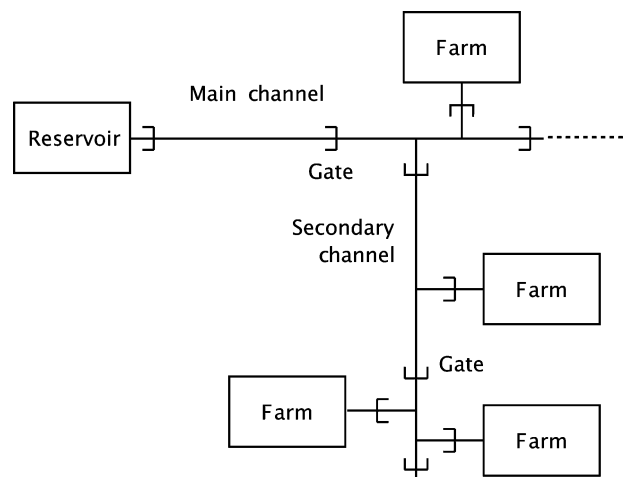


Fig. 1. Top view of irrigation network.

typically no pumping and water is transported along the open-water channels under the power of gravity alone. The flow of water through the network is regulated by gates along the channels, as illustrated in Figs. 2 and 3, where so-called over-shot gates are shown.¹ To give some idea of scale, there are about 7000 km of irrigation channels in the Goulburn-Murray Irrigation District (the largest in Australia), which annually supply more than 2 Gm³ of fresh water for agricultural purposes. And as an example, the relatively small secondary channel “Central-Goulburn No. 2” is 74 km long, with 35 gates and 200 outlets to farms. Irrigation networks exceeding this scale exist in the U.S., Pakistan, India, and China.

¹Although systems with over-shot gates are considered here, the approaches to closed-loop control described herein are not limited to such regulating structures.

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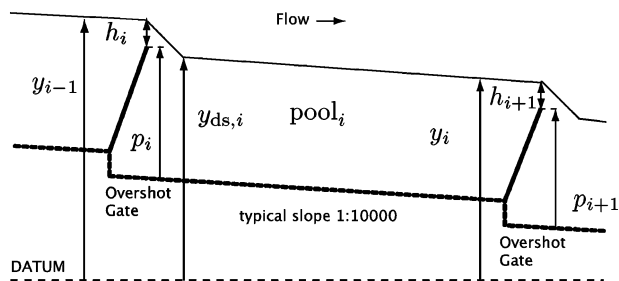


Fig. 2. Stretch of open-water channel with over-shot gates.



Fig. 3. Automated over-shot gates.

The stretch of water between two gates is called a pool. Indeed the open-water channels in an irrigation network can be thought of as strings of pools linked by the regulating gates. The off-take points to farms and secondary channels occur along the pools and the supply of water at these points is also typically powered by gravity. As such, from the perspective of meeting water demands on-farm and within the network, it is important for the water level (i.e., potential energy) in each pool to be maintained above a particular value.

Water losses in irrigation networks can be large, e.g., typically 25% in Australia [2]. Distribution losses are mainly due to oversupply, which can result in spillage along the open channels and irrecoverable outflow at the end of the channels.² From the perspective of distribution efficiency, it is therefore important to go beyond simply ensuring water levels remain above critical values (e.g., by

²Some losses (< 10%) are also due to seepage and evaporation.

oversupplying). In particular, water levels should be maintained at reference set points, calculated on the basis of demand. Furthermore, in terms of achieving quality of service, it is also important to avoid fluctuations from set-point, which translates to fluctuations in flow at off-take points.

In some modern irrigation networks the civil infrastructure of open-water channels and reservoirs is augmented with an information technology infrastructure, comprising sensors and actuators linked through a supervisory control and data acquisition (SCADA) communication network. At each gate, these resources serve to:

- 1) measure local upstream and downstream water levels—i.e., y_{i-1} and $y_{ds,i}$ at gate_{*i*} in Fig. 2;
- 2) electromechanically set the position of the gate—i.e., p_i at gate_{*i*};
- 3) process local information and information communicated over a radio network from other gates and/or a central processing station.

Details of the automated gate system shown in Fig. 3, for example, can be found in [3]. In most fully operational systems, the information technology resources just described are primarily employed to remotely control gate positions, often in open loop, and to acquire data for monitoring and modelling purposes. By the same token, this infrastructure provides ample scope for closed-loop control and the corresponding benefits of robustness to modelling uncertainties. Indeed, recent work strongly suggests that closed-loop control, whereby the gate positions are controlled using information measured online, can lead to significant improvements. See [2], [4]–[15], and the reference therein, for example.

For an irrigation network, a hierarchical controller composed of a supervisory network controller and channel controllers is appropriate. The supervisory controller sits at the upper level of the hierarchy and provides the lower level channel controllers with reference water-level set points, which are computed on the basis of demand and network capacity to avoid the problems associated with oversupply. The role of the channel controllers is to regulate the water levels to the reference set points in the face of the load disturbances associated with off-takes onto farms or into secondary channels. The important problem of supervisory network control is a topic of ongoing investigations and is not considered further here. Rather, we focus on channel control under various information structures for achieving closed-loop set-point regulation and off-take load disturbance rejection.

With reference to Fig. 2, we consider the following information structures for closed-loop control.

- 1) Decentralized feedback—The flow over gate_{*i*} is controlled on the basis of the water level y_i , measured at gate_{*i+1*}, relative to the set point for pool_{*i*}.
- 2) Decentralized feedback with feedforward—The flow over gate_{*i*} is controlled on the basis of the measured water level y_i , relative to the set point for

pool_{*i*}, in addition to information about the flow over gate_{*i+1*}.

- 3) A so-called distributed generalization of the decentralized schemes in which the neighboring decentralized feedback compensators can exchange information.

The focus on these “local” information structures is motivated by the large-scale nature of the open-water channel control problem and the other non-control roles of the underpinning communication infrastructure in the management of an irrigation network.

Note that for an over-shot gate in free flow, as shown in Fig. 3, the flow into the downstream pool is a monotonic function of the head over the gate [16]. In fact, in this case the flow over gate_{*i*} is given by $u_i = \gamma_i h_i^{3/2}$, where γ_i is a constant which depends on the geometry of the gate and h_i is the so-called head over the gate.³ As such, controlling the flow at gate_{*i*} corresponds to adjusting the gate position p_i relative to the upstream water level y_{i-1} .

In the closed-loop channel control schemes just described, the downstream water level in each pool is regulated to a reference set point. The rationale for this is that off-takes typically occur towards the downstream end of each pool and this is where the water level is most important. Furthermore, note that the flow over the upstream gate of a pool is effectively used to regulate the downstream water level. As such, the effect of any control action is always delayed due to the time it takes water to travel from the upstream end to the downstream end of the pool. This delay ultimately limits achievable performance [17], Chs. 8–9. There would be no such delay, on the other hand, if the flow over gate_{*i*} were controlled on the basis of the water level y_{i-1} , immediately upstream. Although an information structure of this kind is useful when very accurate control of an isolated water level is required, the overall performance of such a system tends to be poorer in terms of water wastage when compared to systems with the so-called distant downstream control information structures described previously [11], [18]. The poorer performance in this case is due to the water level being regulated by the flow of water out of the pool, rather than the flow from the source of the water.

In addition to set-point regulation and load disturbance rejection performance, the spatial propagation of water-level errors in the system is also of importance. Under the distant downstream control information structures described, control action taken to compensate for a water-level error in pool_{*i*} affects the flow out of and hence the water level in pool_{*i-1*}.⁴ This, in turn, results in control

action and a corresponding water-level error which continues to propagate upstream, along the string of pools. For channels with a sufficient number of pools, amplification of any component of the water-level error as it propagates upstream can lead to unacceptable deviations from set point, with an associated degradation in the quality of service and even saturation of the gates in the fully open and fully closed positions.

In brief, the paper has the following structure. In Section II, we present a simple model of an irrigation channel. This model is suitable for design under control information structures with feedback paths. The model structure resembles a string of subsystems which each capture the dynamics of the water level at the downstream end of a pool. Importantly, the model is parametrized in a form that is amenable to system identification [19]. Section III serves to discuss the following:

- 1) design of automatic controllers with the two decentralized feedback, with and without feedforward, information structures described previously;
- 2) performance in terms of simulation results;
- 3) analysis of the water-level error propagation characteristics, by which we identify a design tradeoff between local set-point regulation and disturbance rejection performance, on one hand, and water-level error propagation performance, on the other.

Some experimental data, obtained from field tests carried out in collaboration with Rubicon Systems Australia Pty., Ltd., are also presented to demonstrate the validity of the simulations and overall framework. The design and performance of controllers with centralized and distributed control information structures are then considered in Section IV. There, an optimal control approach is proposed for systematically dealing with the design tradeoff identified in Section III. The approach is based on the H_∞ loop-shaping paradigm of McFarlane and Glover [20] and recently developed tools for solving structured optimal control problems [21]–[23]. Supporting simulations are presented. Section V concludes the paper with a brief discussion of related issues not considered explicitly in this paper. In particular, we briefly discuss actuator saturation and antiwindup, fault detection and controller reconfiguration, and performance monitoring.

II. IRRIGATION CHANNEL MODEL FOR CONTROL

Traditionally, open-water channel dynamics are described using the so-called Saint Venant equations [2], [7], [13], [24]–[26]. These nonlinear partial differential equations represent a mass and momentum balance along the length of each pool. Only recently have such models been shown to accurately capture the dynamics of large-scale open-water channels [27]. Within the context

³When not in free flow (i.e., when the immediate downstream water level is above the gate), the flow becomes a function of both the upstream and downstream head over the gate. In this case, the developments that follow continue to hold with suitably modified variables.

⁴Note there is no effect on the flow into the downstream pools.

of control design for set-point regulation and load disturbance rejection, however, the value of this type of model is limited because of its complexity, from both the perspective of system identification and closed-loop analysis and synthesis [2], [9], [15], [19].

The following model of the downstream water level in pool_{*i*}, which is based in principle on conservation of mass and is capable of capturing the first mode of wave phenomena, was recently proposed in [2] and [19]

$$\pi_i\left(\frac{d}{dt}\right)y_i(t) = \gamma_i h_i^{3/2}(t - \tau_i) - \gamma_{i+1} h_{i+1}^{3/2}(t) - d_i(t). \quad (1)$$

With reference to Fig. 2, the terms $h_i^{3/2}$ and $h_{i+1}^{3/2}$ are proportional to the flow over gate_{*i*} and gate_{*i+1*}, respectively, d_i models water off-take disturbances and τ_i is the delay associated with the time it takes for the water to travel the length of the pool. The model dynamics are characterized by the polynomial $\pi_i(\cdot)$. For pools of length up to several kilometers, high-fidelity models can be obtained using a third-order polynomial with roots corresponding to an integrator and an under damped oscillatory mode. The integrator captures the mass balance and the oscillatory mode represents the dynamics of the dominant wave behavior [2], [19]. The polynomial $\pi_i(\cdot)$ and other model parameters γ_i , γ_{i+1} , and τ_i in (1) are readily obtained via standard linear system identification techniques, using either open-loop data or closed-loop data [19], [27], [28]. Importantly, a model of each pool can be identified from

locally measured information. Fig. 4 shows a measured water level (solid) and the simulated water level (dashed-dotted) obtained with a third-order model identified from experimental data. For a more detailed discussion of the simplified model proposed here, see [2] and [19], and for a detailed comparison with a corresponding Saint Venant equations-based model see [27].

Since the free space above the typical set point in an open-water channel is not designed to cope with waves, an additional control objective is to ensure that the wave dynamics are not excited. Waves are also undesirable because they lead to fluctuations in the flow at off-take points onto farms and into secondary channels. As such, care must be taken to ensure that each local control-loop bandwidth lies at a frequency below the corresponding dominant wave phenomenon. Provided this is taken into consideration during the design of feedback controllers, the first-order model

$$\alpha_i \frac{d}{dt} y_i(t) = \gamma_i h_i^{3/2}(t - \tau_i) - \gamma_{i+1} h_{i+1}^{3/2}(t) - d_i(t) \quad (2)$$

where α_i is a measure of pool surface area, is adequate because it captures the dynamics of the water levels from the perspective of the required closed-loop behavior. Indeed, a model of this kind was employed for feedback controller design in [6], [9], [10], [14], and [29], for example.

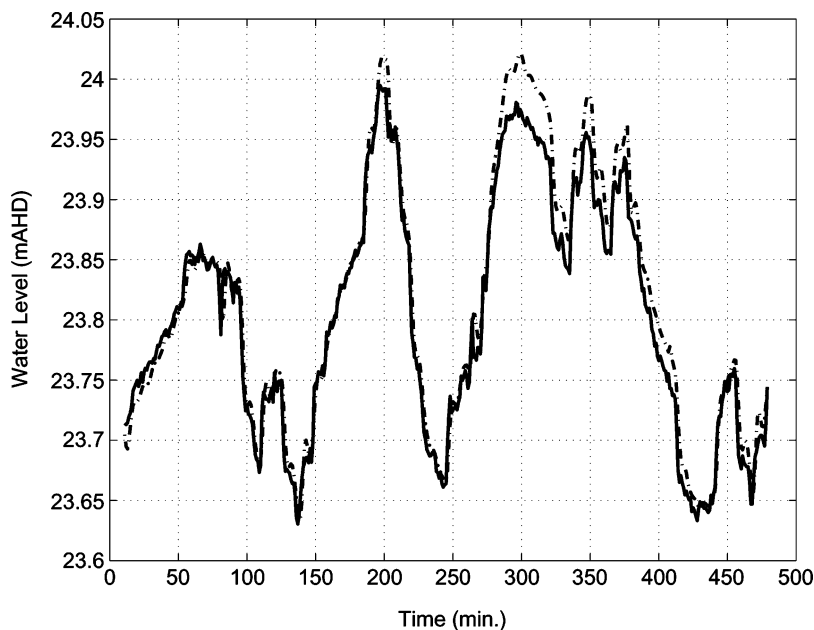


Fig. 4. Validation of third-order model of form (1).

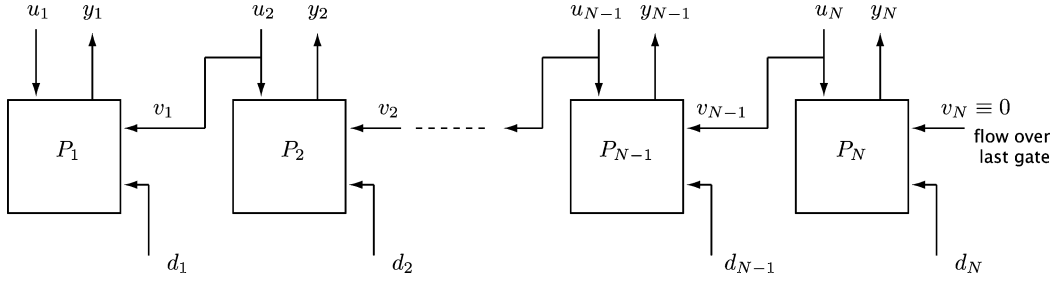


Fig. 5. Irrigation channel modelled as a string of N pools with dynamics described by (3).

Linearizing via the change of variables

$$u_i := \gamma_i h_i^{3/2}$$

which is a measure of the flow over gate _{i} , and applying the Laplace transform, yields the frequency domain model

$$P_i : y_i(s) = \frac{1}{s\alpha_i} (\exp(-s\tau_i) u_i(s) - v_i(s) - d_i(s)) \quad (3)$$

where

$$v_i(s) := u_{i+1}(s)$$

with the boundary condition $v_N \equiv 0$,⁵ and for ease of notation no distinction is made between the time-domain and frequency-domain representations of a signal. As a whole, the irrigation channel model can be thought of as a string of pool models (3), as illustrated in Fig. 5. This is the model employed in the subsequent sections.

III. DECENTRALIZED CONTROL

The two decentralized control information structures described in Section I (with and without feedforward paths) are considered here from the perspective of achieving set-point regulation of water levels and off-take load disturbance rejection, while avoiding the excitation of wave dynamics. In particular, the design of controllers with these information structures, and the relative merits of each, are discussed in terms of the control objectives just mentioned. Some experimental data, obtained from field trials carried out in collaboration with Rubicon Systems Australia Pty.,

Ltd., are presented to demonstrate the validity of the overall design framework. Analysis of the water-level error propagation characteristics is also carried out. The analysis, which is similar in spirit to the analysis of controlled vehicle strings in [30], identifies a tradeoff between local set-point regulation and disturbance rejection performance, on one hand, and water-level error propagation performance, on the other. This motivates subsequent consideration of centralized and distributed generalizations of the decentralized schemes considered here.

A. Decentralized Feedback

The decentralized feedback control information structure described in Section I is illustrated in Fig. 6. Note that

$$u_i(s) = C_i(s)e_i(s) := C_i(s)(r_i(s) - y_i(s)).$$

As such, the control action at gate _{i} is determined on the basis of the local water-level set-point r_i , and the measured water level y_i , which is communicated from gate _{$i+1$} . As in [9], [10], and [15], the design of each feedback compensator C_i can be carried out from the perspective of achieving good local set-point regulation and disturbance rejection. This corresponds to thinking of v_i in (3) as being part of the disturbance and focussing on the local closed-loop transfer functions

$$T_{r_i \rightarrow e_i}(s) := \frac{1}{1 + L_i(s) \exp(-s\tau_i)}$$

$$T_{d_i \rightarrow e_i}(s) := \frac{1}{1 + L_i(s) \exp(-s\tau_i)} \left(\frac{1}{s\alpha_i} \right)$$

and

$$T_{d_i \rightarrow u_i}(s) := \frac{L_i(s)}{1 + L_i(s) \exp(-s\tau_i)}$$

where $L_i(s) := C_i(s)/s\alpha_i$.

⁵That is, no flow over the last gate. Provided the water level y_N remains sufficiently close to set point, the flow over the last gate can always be independently controlled to be zero. This is important since water which passes the last gate is usually wasted.

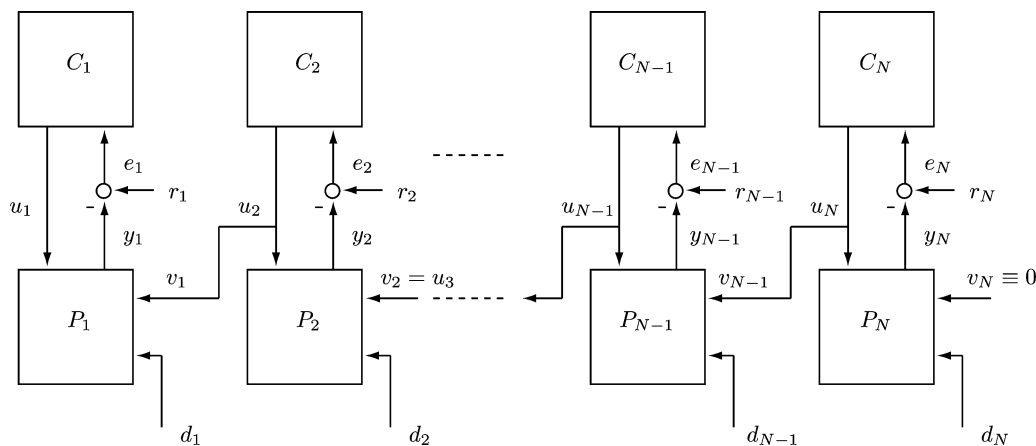


Fig. 6. Decentralized feedback control information structure.

By the final value theorem, $C_i(s)$ must have at least one pole at $s = 0$ in order to achieve zero steady-state water-level error for step load disturbances d_i . Furthermore, in line with classical design ideas [17], [31]–[33], good set-point regulation and load disturbance rejection is achieved by ensuring that the local loop-gain $|L_i(j\omega)|$ is large at frequencies where the set-point reference r_i and the load disturbance d_i are significant (i.e., low frequency). Moreover, the bandwidth of the loop gain $|L_i(j\omega)|$ must be set to lie below the frequency of the local wave dynamics, which are not captured by the first-order model (3). Indeed, these local control objectives can be achieved with an essentially PI compensator

$$C_i(s) = \frac{\kappa_i(1 + s\phi_i)}{s(1 + s\rho_i)}$$

which also introduces phase lead at the desired loop gain bandwidth for stability,⁶ robustness, and to accommodate the delay, as determined by the constants κ_i and ϕ_i . The compensator also introduces additional rolloff beyond this to ensure low gain at the dominant wave frequency, as determined by the constant ρ_i . Fig. 7 shows tuned local loop gains $|L_i(j\omega) := C_i(j\omega)/j\omega\alpha_i|$ for a string of five pool models with delay parameters and length parameters set to the values summarized in Table 1. Observe that the bandwidth of each loop gain has been set relative to the dominant wave frequencies φ_i in each pool; see Table 1.

Simulations of the corresponding closed-loop behavior of the water levels are shown in Fig. 8. For each pool, the

third-order model

$$\pi_i \left(\frac{d}{dt} \right) y_i(t) = u_i(t - \tau_i) - u_{i+1}(t) - d_i(t) \quad (4)$$

is used, where $\pi_i(\xi) = \alpha_i \xi (\xi^2 + 2\zeta_i \omega_{n,i} \xi + \omega_{n,i}^2) / \omega_{n,i}^2$ and the damping ratios and natural frequencies satisfy $\varphi_i = \omega_{n,i} \sqrt{1 - \zeta_i^2}$. Water levels are shown in mAHD, the height in meters above a standardized reference datum. The simulation scenario corresponds to an off-take disturbance in pool₃, starting just before 0 min and finishing at 600 min, an off-take disturbance in pool₅, starting at 1560 min and ending at 3000 min, and a water-level set-point change in pool₃ of 0.1 m at 4820 min. This corresponds (approximately) to a 4000 m³ steady-state change in the volume of water in pool₃. The decentralized feedback controller achieves good local set-point regulation and load disturbance rejection. See, in particular, y_3 after 4820 min and y_5 between 1560 and 4000 min. On the other hand, upstream propagation of the water-level error induced in pool₅ at 1560 min is also evident. Indeed, it would appear that a component of the water-level error is amplified as it propagates upstream.

To understand the error-propagation phenomenon observed, consider for the sake of argument a string of *identical* pools, each with delay τ and suppose the same decentralized feedback compensator $C(s)$ is used for each pool. In this case, the transfer functions from the set-point reference and disturbance to the water-level error are given by

$$T_{r_i \rightarrow e_i}(s) := \frac{1}{1 + L(s) \exp(-s\tau)} = s\alpha T_{d_i \rightarrow e_i}(s)$$

⁶There will be a double integrator in the loop.

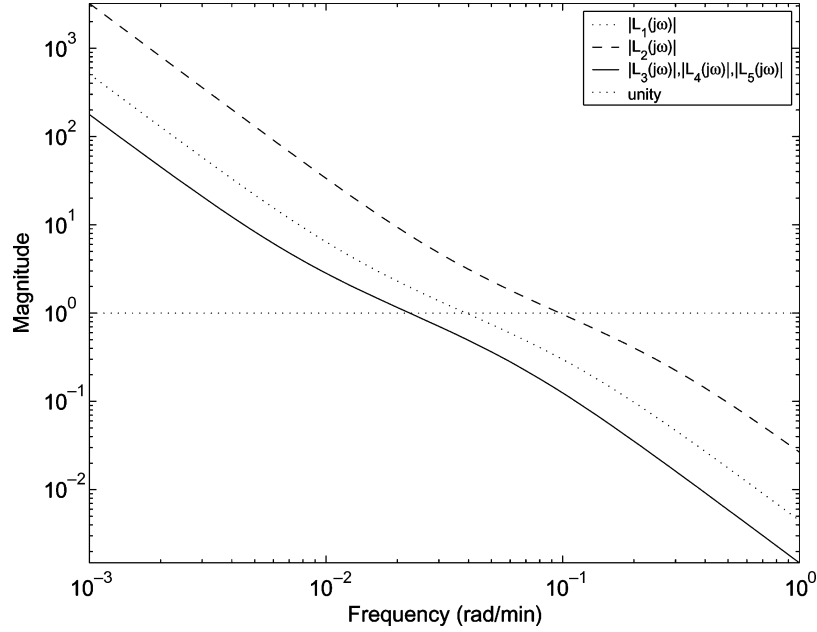


Fig. 7. Tuned decentralized loop-gains $|L_i|$.

and the water-level error propagates according to the transfer function

$$T_{e_{i+1} \rightarrow e_i}(s) := \frac{L(s)}{1 + L(s) \exp(-s\tau)}$$

where $L(s) := C(s)/s\alpha$ and α is a measure of the surface area of each pool. This is intuitively a worst case scenario because the local loop-gain bandwidths in neighboring pools are identical in this case; hence, there can be significant coupling of control action into upstream pools. Observe, in particular, that

$$T_{r_i \rightarrow e_i}(s) + T_{e_{i+1} \rightarrow e_i}(s) \exp(-s\tau) = 1$$

and, thus, that there is a tradeoff between set-point regulation and load disturbance rejection performance, on one hand, and water-level error propagation performance, on the other.

Table 1 Delay, Surface Area, and Wave Frequency Parameters Extrapolated From Validated System Identification Models

i	τ_i	α_i	φ_i
1	8 mins	22414m ²	0.42 rad/min
2	3 mins	11942m ²	0.74 rad/min
3, 4, 5	16 mins	43806m ²	0.20 rad/min

Fig. 9 shows a simulation of the deviation from set point of the closed-loop water levels in the three identical pools pool₃, pool₄ and pool₅, for an off-take disturbance in pool₅. The corresponding third-order model (see (4) and Table 1) is used for the simulation and each decentralized feedback compensator is taken to be

$$C(s) = \frac{\kappa(1 + s\phi)}{s(1 + s\rho)} \quad (5)$$

with $\kappa = 7.72$, $\phi = 128$, and $\rho = 15.2$. The local disturbance rejection performance is good—see water-level error y_5 . However, it can be seen that a component of the water-level error in pool₅ is being amplified as it propagates upstream—also see water-level errors $e_1, \dots, e_4$ in Fig. 9. In fact, the following analytical result, which was recently observed in [34], implies water-level errors are amplified with any linear decentralized compensator $C(s)$ that achieves zero steady-state error in the face of step load disturbances.

Lemma 1: Let $C(s)$ be any proper rational transfer function such that $C(s)$ has a pole at $s = 0$ and the local closed-loop transfer function

$$T_{r_i \rightarrow e_i}(s) := \frac{1}{1 + L(s) \exp(-s\tau)}$$

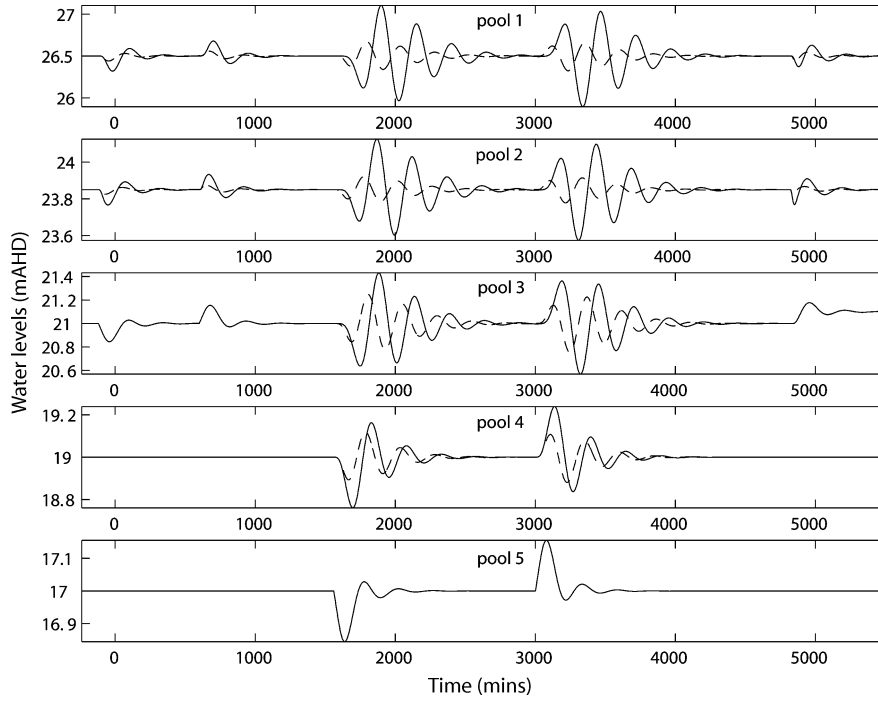


Fig. 8. Closed-loop water levels under decentralized feedback control information structures (solid—without feedforward; dashed—with feedforward).

is stable, where $L(s) := C(s)/s\alpha$. Then, there exists an $\hat{\omega} \in [0, \infty)$ such that $|T_{e_{i+1} \rightarrow e_i}(j\hat{\omega})| > 1$, where

$$T_{e_{i+1} \rightarrow e_i}(s) := \frac{L(s)}{1 + L(s) \exp(-s\tau)}.$$

Proof: As described in [17], ch. 9, applying Cauchy's theorem to

$$\int_{\mathcal{D}} \frac{1}{s^2} T_{e_{i+1} \rightarrow e_i}(s) ds$$

where $\mathcal{D}$ is the standard Nyquist contour with an indentation around the origin in the complex plane, yields

$$\int_0^\infty \ln |T_{e_{i+1} \rightarrow e_i}(j\omega)| \frac{d\omega}{\omega^2} \geq 0.$$

Since $T_{e_{i+1} \rightarrow e_i}(s)$ is strictly proper, this implies the existence of a frequency $\hat{\omega}$ such that $\ln |T_{e_{i+1} \rightarrow e_i}(j\hat{\omega})| > 0$. ■

A detailed analysis of the error propagation characteristics for the case of a string comprising different pools

can be found in [35], where it is shown that the ratio of neighboring compensators has a direct influence on how the error propagates.

B. Decentralized Feedback With Feedforward

The water-level error amplification phenomenon described can be a significant problem. It can lead to saturation of the regulating gates and deviations from set points which are large enough to cause unacceptable fluctuations in the flows at off-take points and even flooding. Towards managing this phenomenon, the propagation of water-level errors can be thought of as the propagation of disturbances on the flow over the downstream gate of each pool. These flow disturbances correspond to control actions, hence a measure of this disturbance can be made available for control at the upstream gate, without additional cost in terms of the required communication infrastructure. This is precisely the decentralized feedback with feedforward control information structure described in the introduction; also see Fig. 10. Note, in particular, that the flow over gate_{*i*} is now controlled according to

$$u_i(s) = C_i(s)e_i(s) + F_i(s)u_{i+1}(s)$$

where each feedforward compensator $F_i(s)$ is a stable transfer function designed to manage the error propagation characteristics of the closed loop, given decentralized

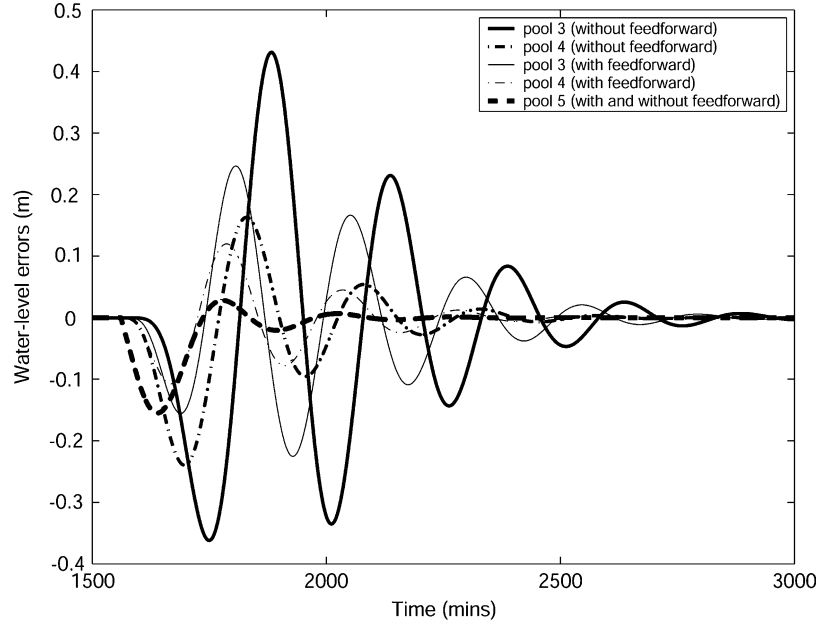


Fig. 9. Water-level error amplification in identical pools under decentralized feedback control information structures.

feedback compensators $C_i(s)$ designed, as described, to achieve good local set-point regulation and load disturbance rejection.

Again, for the sake of argument, consider a string of identical pools, each with delay τ . Now suppose that the same feedback compensator $C(s)$, of the form (5), and feedforward compensator $F(s)$, are used for each pool. Then, the water-level error propagation characteristics are governed by the transfer function

$$T_{v_i \rightarrow e_i} := \frac{1}{s\alpha} \left(\frac{1 - F(s) \exp(-s\tau)}{1 + L(s) \exp(-s\tau)} \right)$$

from v_i to e_i in each pool and the relationship

$$v_i(s) = C(s)e_{i+1}(s) + F(s)v_{i+1}(s)$$

where $L(s) := C(s)/s\alpha$ and α is a measure of the surface area of each pool. Combining these yields the following relationship between the water-level error in neighboring pools:

$$e_i(s) = T_{e_{i+1} \rightarrow e_i}(s)e_{i+1}(s)$$

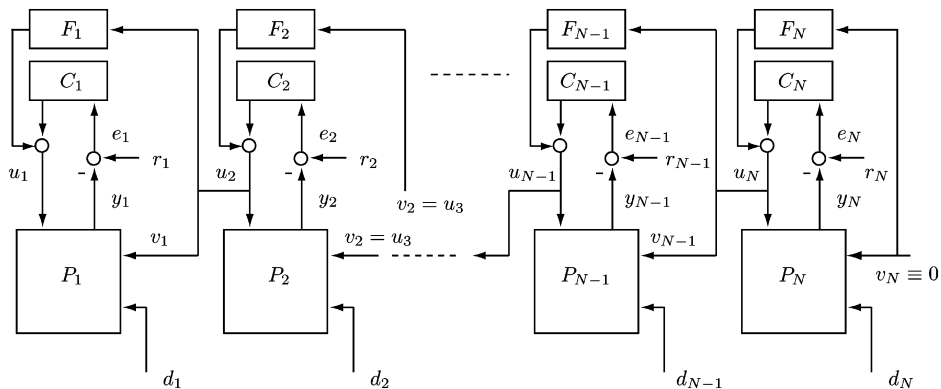


Fig. 10. Decentralized feedback with feedforward control information structure.

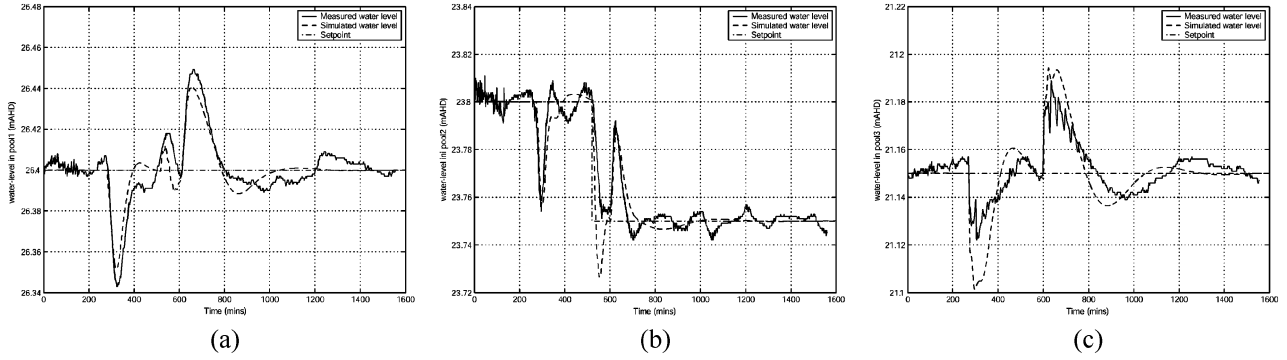


Fig. 11. Measured closed-loop water levels under decentralized feedback control with feedforward. Field test results on Houghton main channel. (a) Water level in pool₁. (b) Water level in pool₂. (c) Water level in pool₃.

where

$$T_{e_{i+1} \rightarrow e_i}(s) := \frac{L(s) + F(s)}{1 + L(s) \exp(-s\tau)}.$$

Observe that by augmenting the decentralized feedback compensators with the decentralized feedforward compensators $F(s)$, an extra degree of freedom is gained in terms of managing the error propagation characteristics. In the simulations shown in Figs. 8 and 9, it can be seen that the inclusion of the feedforward (or decoupling) paths can notably improve the error-propagation characteristics of the controlled system.

Fully exploiting the additional freedom offered by the feedforward paths, however, appears to be difficult. In particular, for the decentralized feedback compensator (5), which can be designed to achieve good local set-point regulation and disturbance rejection performance as described, and under the constraint that the open-loop transfer function $F(s)$ be proper and stable, it follows that $|T_{e_{i+1} \rightarrow e_i}(0)| = 1$. Moreover, by observing that at sufficiently low frequencies $L(j\omega) \approx (-\mu/\omega^2)$ with $\mu > 0$, it can be shown that

$$\frac{d}{d\omega} |T_{e_{i+1} \rightarrow e_i}(j\omega)| > 0$$

for $\omega \in [0, \omega_0)$ and an $\omega_0 > 0$. Hence, as observed before without the feedforward paths, there always exists a frequency $\hat{\omega}$ such that $|T_{e_{i+1} \rightarrow e_i}(j\hat{\omega})| > 1$. Furthermore, in order to achieve a reduction in $|T_{e_{i+1} \rightarrow e_i}(j\omega)|$ where $|L(j\omega)|$ is large (e.g., within the loop-gain bandwidth), $F(j\omega)$ must be of comparably large magnitude. In this case, the gain from the flow disturbance v_i , resulting from control action taken in pool _{$i+1$} , to the water-level error e_i is large, since $|T_{v_i \rightarrow e_i}(j\omega)|$ behaves like $1/\omega\alpha$. Indeed, experience suggests that the best one can do is ensure that $|T_{v_i \rightarrow e_i}(j\omega)|$ is rea-

sonably small within the bandwidth of the loop-gain $|L(j\omega)|$, while ensuring that $|T_{e_{i+1} \rightarrow e_i}(j\omega)|$ is not much greater than one at any frequency.

C. Field Tests

The models used for pools 1, 2, and 3 in the simulations are extrapolated from models identified using data obtained for three successive pools of the Houghton Main Channel, Queensland Australia.⁷ The identified models were used to design decentralized feedback and feedforward compensators, as described. These were subsequently tested in the field. Fig. 11 shows the field test results. The experiment scenario corresponds to a disturbance in pool₃, which is induced by moving the downstream gate at around 250 and 600 min, and a set-point change in pool₂ at around 550 min. Simulated results, based on the corresponding third-order pool model of the form (1), are also shown. It can be seen that the local disturbance rejection and set-point regulation performance is excellent, with water levels recovering smoothly. The upstream propagation of the water-level error induced in pool₃ is also evident. Fig. 12 shows the improvement achieved by including the feedforward compensators. The slow oscillation around set point which can be seen most clearly in pool₂ is due to the use of a 0.015-m dead band on gate position movement in order to limit any potential wearing of the gates. Finally, note that the measured data agree well with the simulated values. More details are available in [9] and [15], where the pools considered here are referred to as pools 8, 9, and 10.

IV. OPTIMAL CENTRALIZED AND DISTRIBUTED CONTROL

In this section, an optimal control approach is proposed for systematically managing the tradeoff, identified in Section III, between the set-point regulation and load

⁷The two extra pools 4 and 5 are added to the simulations to more clearly illustrate the water-level error propagation phenomenon.

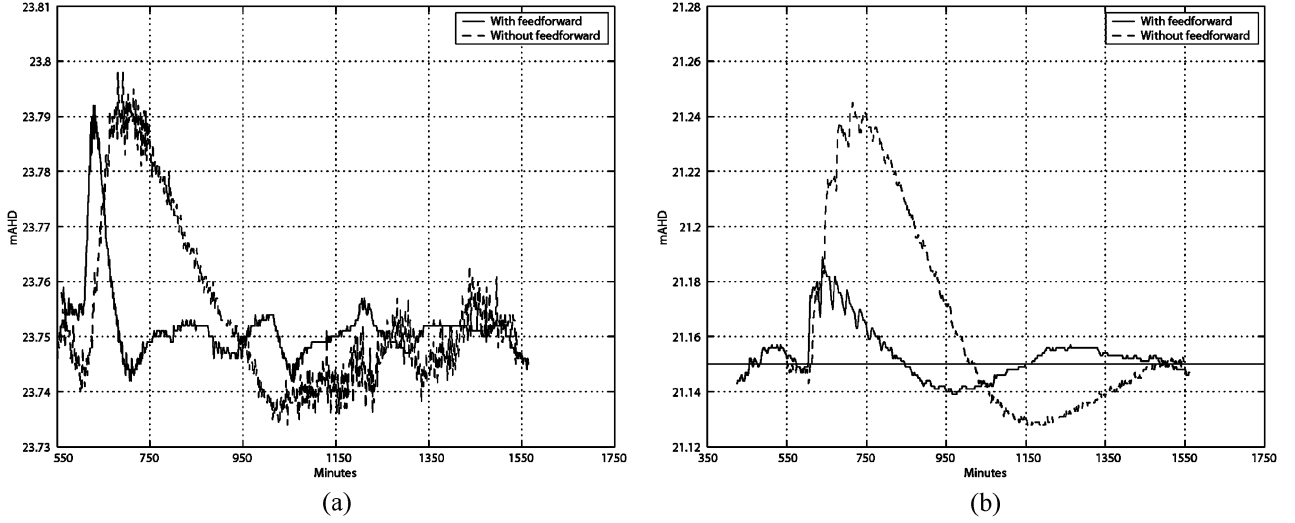


Fig. 12. Measured closed-loop water levels under decentralized feedback control with and without feedforward, including that of disturbance induced in pool₃. Field test results on Houghton main channel. (a) Water level in pool₂. (b) Water level in pool₃.

disturbance rejection performance of the controlled system and the error propagation performance. The approach is based on the H_∞ loop-shaping paradigm of McFarlane and Glover [20]. Both a centralized and distributed generalization of the decentralized control information structures described in Section III are considered. Although the centralized control formulation is not ideally suited to the large-scale problem at hand, it is considered to give a feel for what is achievable when all water levels are available for the control of each water level and to facilitate assessment of the distributed control information structure.

A. Centralized Control

The centralized control information structure described in the introduction is illustrated in Fig. 13.

Towards designing the centralized (MIMO) feedback compensator shown, the string of pools can be modelled as a whole with the MIMO transfer function

$$P(s) := \begin{pmatrix} \frac{\exp(-s\tau_1)}{s\alpha_1} & \frac{-1}{s\alpha_1} & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \frac{-1}{s\alpha_{N-1}} \\ 0 & \dots & \dots & 0 & \frac{\exp(-s\tau_N)}{s\alpha_N} \end{pmatrix} \quad (6)$$

between the vector of flows $u := (u_1, \dots, u_N)^T$ and the vector of water levels $y := (y_1, \dots, y_N)^T$. This follows

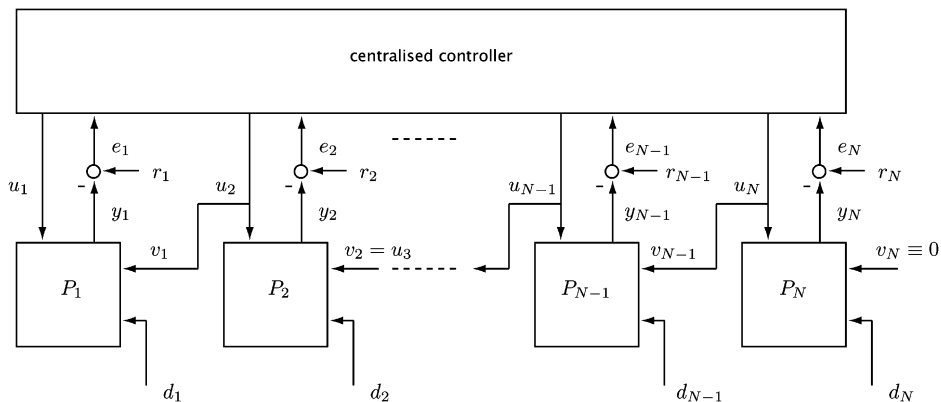


Fig. 13. Centralized control information structure.

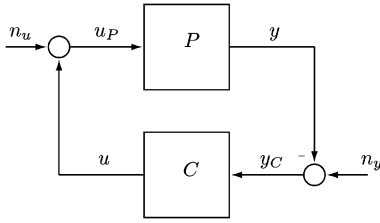


Fig. 14. Standard feedback interconnection.

readily from the parametrized model (3) used for each pool in Section III. Then, with $C(s)$ denoting the transfer function of the (MIMO) centralized controller, the closed-loop system shown in Fig. 13 corresponds to the standard feedback interconnection structure shown in Fig. 14, with

$$n_y(s) = r(s) + \text{diag}\left(\frac{1}{s\alpha_i}\right)d(s)$$

$y_C(s) = e(s) := (e_1, \dots, e_N)^T$ and $n_u \equiv 0$, where $r := (r_1, \dots, r_N)^T$ is the vector of water-level set points and $d := (d_1, \dots, d_N)^T$ is the vector of off-take disturbances.

Within the H_∞ loop-shaping paradigm for design [20], [32], [33], [36], robustness to uncertainty and performance objectives, such as set-point regulation and disturbance rejection, are achieved with a feedback compensator of the form $C(s) = W(s)C_\infty(s)$. In particular:

- 1) $W(s)$ is a proper rational transfer function weight⁸ designed so that the singular values of $P(j\omega)W(j\omega)$ exhibit classical loop-shapes corresponding to the performance and robustness objectives across frequency;
- 2) the optimal controller

$$C_\infty(s) := \underset{C_{\text{syn}} \in \mathcal{C}}{\text{argmin}} \|H(P(s)W(s), C_{\text{syn}}(s))\|_\infty \quad (7)$$

where

$$H(X, Y) := \begin{pmatrix} Y \\ I \end{pmatrix} (I + XY)^{-1} \begin{pmatrix} -X & I \end{pmatrix},$$

$$\mathcal{C} := \{C_{\text{syn}} : H(P(s)W(s), C_{\text{syn}}(s)) \in H_\infty\},$$

⁸Typically restricted to be minimum phase and at least marginally stable.

H_∞ denotes the standard Hardy space of stable transfer functions with norm⁹ defined by

$$\|H\|_\infty := \max_\omega \bar{\sigma}(H(j\omega))$$

and $\bar{\sigma}(\cdot)$ denotes the maximum singular value of the matrix argument.¹⁰

Observe that the transfer function $H(PW, C_{\text{syn}})$ in (7) is a weighted version of the closed-loop mapping

$$n := \begin{pmatrix} n_u \\ n_y \end{pmatrix} \mapsto \begin{pmatrix} u \\ y_C \end{pmatrix} =: z$$

in Fig. 14. In particular, for the irrigation channel model $P(s)$ in (6), the corresponding cost in (7) penalizes:

- 1) the effect of perturbations to the flow vector u on the water-level error vector e (due to control actions for example), which is characterized by the transfer function $n_u \mapsto y_C$;
- 2) the effect of set-point changes and load disturbances on the water-level errors, which is characterized by the transfer function $n_y \mapsto y_C$;
- 3) the effect of these disturbances on the flow, i.e., u .

This is in direct line with the control objectives identified in the preceding sections, and, importantly, it captures performance from the perspective of set-point regulation, load disturbance rejection, and water-level error propagation. Moreover, the cost is related to a measure of closed-loop robustness [32], [36], which is important since the model $P(s)$ is constructed from only first-order models of each pool. The optimal cost is always greater than or equal to one and a value of around three or four is usually a good indicator of a sensible design. The closer it is to one though, the better the robustness and performance, with respect to the weight $W(s)$.

In Section III, the approach to designing decentralized feedback compensators $C_i(s)$ actually corresponds to the design of a transfer function weight $W(s) = \text{diag}(C_i(s))$ so that the singular values of $\text{diag}(1/s\alpha_i)W(s) = \text{diag}(L_i(s))$ exhibit classical loop-gain shapes; see Fig. 7. In fact, this diagonal weight can be used to obtain the centralized compensator $C(s) = W(s)C_\infty(s)$, via (7), provided the singular values of $P(s)W(s)$ are not too different (from a closed-loop perspective) in shape from the local loop gains $|L_i(j\omega)|$. Experience suggests that this is the case if reasonable performance was achieved by the decentralized compensators; see Fig. 15. By virtue of the cost in (7), the compensator $C_\infty(s)$ offers some robustness to the coupling between pools which is neglected in the process just

⁹This norm corresponds to the energy gain of the system.

¹⁰Standard software tools are available for solving this optimal control problem when the model $P(s)$ is finite dimensional. To this end, we employ a Padé approximation of the time-delay terms in the model (6).

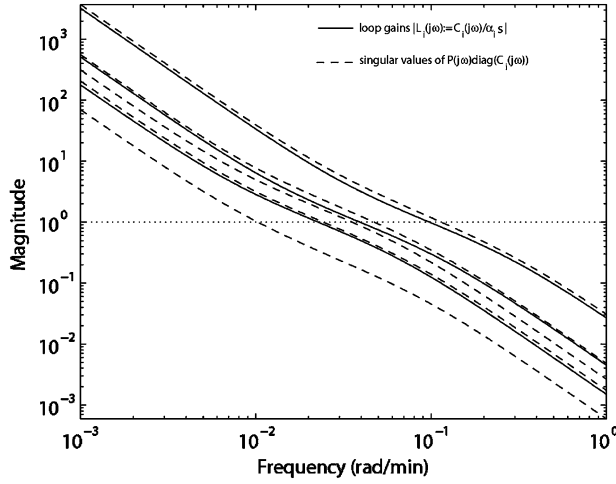


Fig. 15. Decentralized loop gains and singular values of $P(j\omega)W(j\omega)$.

described for obtaining the weight $W(s) = \text{diag}(C_i(s))$, while ensuring that the closed-loop transfer functions governing performance are appropriately penalized across frequency. For more details on the robustness of centralized loop-shaping compensation for irrigation channels, see [29], [35], and [37].

Fig. 16 shows a simulation of the closed-loop water levels for the string of five pools corresponding to the third-order models presented in Section III, under a centralized control information structure for which the controller $C(s)$ was designed as just described.¹¹ Observe that the centralized controller achieves significantly better water-level error propagation performance than the decentralized feedback control controller with feedforward. This appears to be at the expense of a less remarkable degradation in local set-point regulation and load disturbance rejection performance. In effect, the centralized controller achieves a tradeoff between these competing performance objectives by taking less aggressive control action to compensate for water-level errors, compared to the decentralized feedback controller with feedforward. If better local performance is required, the weight $W(s) := \text{diag}(C_i(s))$ suggested can be adjusted accordingly, often without seriously degrading the resultant error propagation characteristics; see Fig. 17.

In contrast with LQ-based techniques [38], the approach described appears to be scalable from the perspective of designing the optimization weight $W(s)$. However, the computational cost associated with solving the optimal control problem and the communication infrastructure cost for centralized control does not scale well, as the number of pools is increased. In the following section, we discuss the application of the H_∞ loop-shaping ideas just

¹¹That is, using the H_∞ loop-shaping procedure described, with a weight $W(s) = \text{diag}(C_i(s))$, where $C_i(s)$ is the decentralized compensator that achieves the loop-gain $|L_i(j\omega)| := C_i(s)/j\omega\alpha_i$ shown in Fig. 7.

described under a so-called distributed generalization of the decentralized control information structures of Section III. Under the distributed control information structure neighboring local compensators are able to exchange information without increasing the communication infrastructure costs associated with decentralized control. The optimal synthesis step requires recently developed tools for solving structured optimal control problems parametrized in terms of state-space realizations of the systems involved [21]–[23].

B. Distributed Control

Fig. 18 illustrates the distributed generalization of the decentralized feedback with feedforward control information structure of Fig. 10. Here, each reference set-point r_i and load disturbance d_i have been lumped into the signal $n_i = (n_{y,i}, n_{u,i})^T := (r_i, d_i, n_{u,i})^T$, where $n_{u,i}$ models an additional disturbance on the flow at gate_{*i*}. The transfer function of each generalized pool is given by

$$G_i(s) := \begin{pmatrix} v_i(s) \\ n_i(s) \\ u_i(s) \end{pmatrix} \mapsto \begin{pmatrix} w_i(s) \\ z_i(s) \\ e_i(s) \end{pmatrix} = \begin{pmatrix} 0 & (0 \ 0 \ 1) & 1 \\ \left(\frac{1}{s\alpha_i}\right) & \begin{pmatrix} 1 & \frac{1}{s\alpha_i} & \frac{-\exp(-s\tau_i)}{s\alpha_i} \\ 0 & 0 & 0 \end{pmatrix} & \begin{pmatrix} \frac{-\exp(-s\tau_i)}{s\alpha_i} \\ 1 \end{pmatrix} \\ \frac{1}{s\alpha_i} & \begin{pmatrix} 1 & \frac{1}{s\alpha_i} & \frac{-\exp(-s\tau_i)}{s\alpha_i} \end{pmatrix} & \frac{-\exp(-s\tau_i)}{s\alpha_i} \end{pmatrix} \quad (8)$$

and the interconnection satisfies the constraint

$$v_i = w_{i+1}$$

with boundary condition $v_N \equiv 0$. Accordingly, the control objectives previously discussed correspond to managing the collective effect of the exogenous signals $n_i := (r_i, d_i, n_{u,i})^T$ on the signals

$$z_i = \begin{pmatrix} e_i \\ u_i \end{pmatrix} := \begin{pmatrix} r_i - y_i \\ u_i \end{pmatrix}.$$

This is to be achieved with the distributed compensators $K_i = \begin{pmatrix} v_i^k \\ e_i \end{pmatrix} \mapsto \begin{pmatrix} w_i^k \\ u_i \end{pmatrix}$, under the interconnection constraint

$$v_i^k = w_{i+1}^k$$

with the boundary condition $v_N^k \equiv 0$.

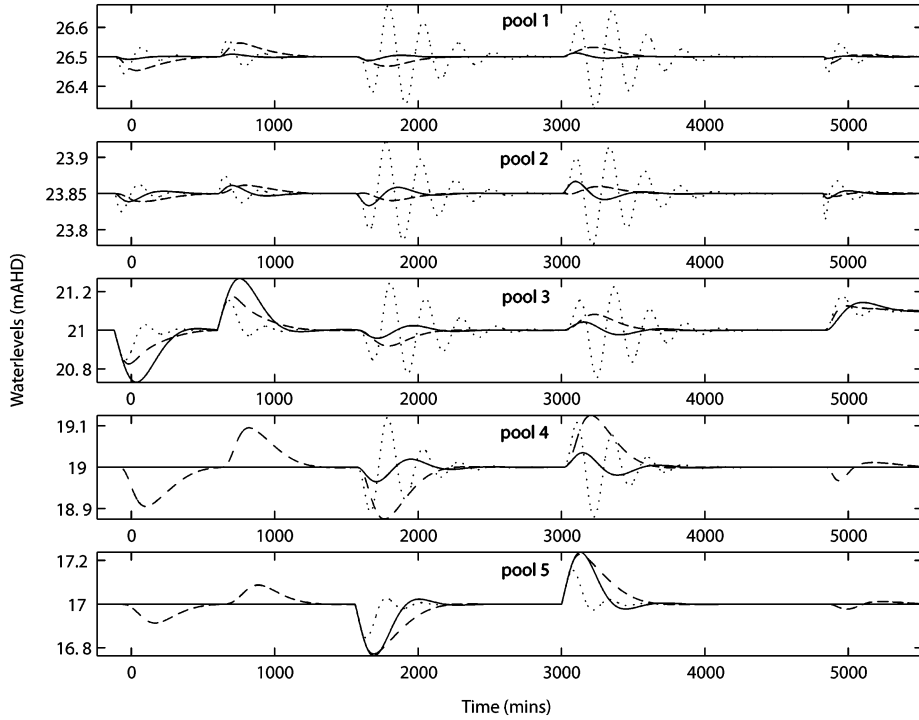


Fig. 16. Closed-loop water levels under decentralized feedback with feedforward, centralized, and distributed control information structures (solid—distributed control; dashed—centralized control; dotted—decentralized feedback with feedforward control).

For convenience, let the overall closed-loop transfer function from the vector of disturbances $n(s) := (n_1, \dots, n_N)^T$ to the vector of penalized signals $z(s) := (z_1, \dots, z_N)^T$ is denoted by

$$H(G, K) := n(s) \mapsto z(s)$$

where $G := (G_1, \dots, G_N)$ and $K := (K_1, \dots, K_N)$.

In line with the centralized H_∞ loop-shaping approach, one would expect the closed-loop robustness and performance objectives described previously to be achievable with distributed compensators

$$K_i(s) = \begin{pmatrix} v_i^K(s) \\ e_i(s) \end{pmatrix} \mapsto \begin{pmatrix} w_i^K(s) \\ u_i(s) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & C_i(s) \end{pmatrix} K_{\infty,i}(s)$$

where:

- 1) $C_i(s)$ is the decentralized feedback compensator that achieves the desired local loop-gain shape $|L_i(j\omega)| = |C_i(j\omega)/j\omega\alpha_i|$, as described in Section III;

- 2) the optimal distributed controllers

$$(K_{\infty,1}, \dots, K_{\infty,N}) := K_\infty := \operatorname{argmin}_{K_{\text{syn}} \in \mathcal{K}} \|H(G_W, K_{\text{syn}})\|_\infty,$$

where $\mathcal{K}$ is the set of all stabilizing distributed compensators for the weighted generalized plant

$$G_W(s) := (G_1(s), \dots, G_N(s)) \operatorname{diag} \begin{pmatrix} I_{3 \times 3} & 0 \\ 0 & C_i(s) I_{2 \times 2} \end{pmatrix}$$

and $I_{n \times n}$ denotes the $n \times n$ identity matrix.

The optimal control problem involved here is simply a structured version of the centralized optimal problem considered previously. In particular, it captures closed-loop robustness and performance objectives in the same way. Furthermore, the problem is amenable to computation via recently established convex programming techniques for solving structured optimal control problems [12], [22], [23].¹²

¹²The techniques generally require the models $G_i(s)$ to be finite dimensional. To this end, we employ a Padé approximation of the time-delay terms in (8).

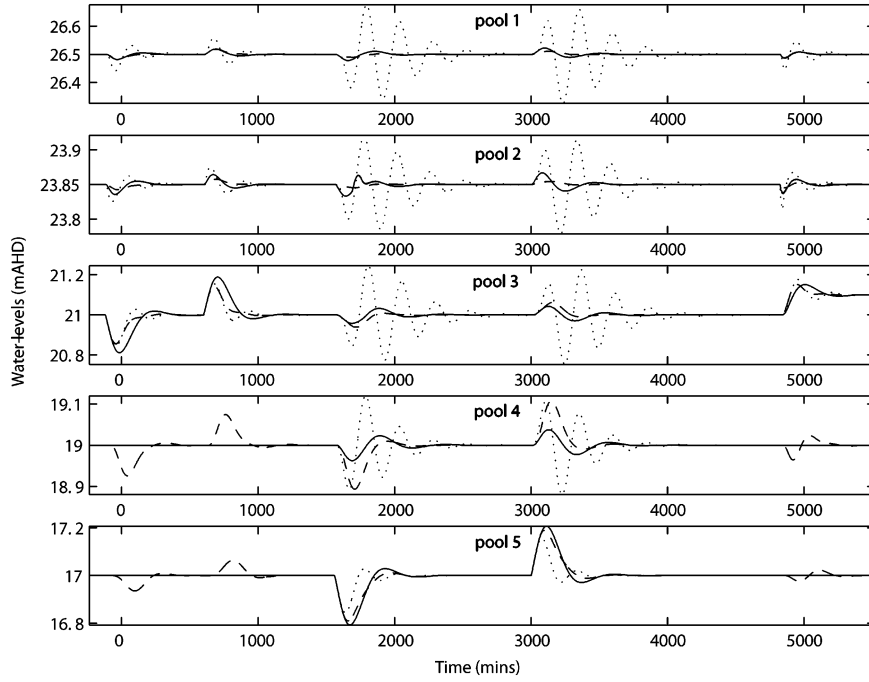


Fig. 17. Closed-loop water levels with tuned centralized and distributed controllers (solid—distributed control; dashed—centralized control; dotted—decentralized feedback with feedforward control).

Fig. 16 includes a simulation of the closed-loop water levels under distributed control with compensators $K_i(s)$ designed as just described and synthesized using the results in [22] and [23]. Observe that the distributed controller achieves similar performance to the centralized controller. Again, the controller appears to achieve an acceptable tradeoff between error propagation performance, on one hand, and local set point and load disturbance rejection performance, on the other. Again, this appears to be the result of less aggressive control action when compensating

for water-level errors, compared to the decentralized feedback controller with feedforward.

Note that the design effort is no greater than that described in Section III for the decentralized control information structures, since the weighting matrix involved can be initially taken to be a diagonal matrix constructed from the decentralized feedback compensators. Furthermore, it is possible to tune the weighted generalized plant G_W to achieve better local performance, without significantly degrading the error propagation characteristics; see Fig. 17.

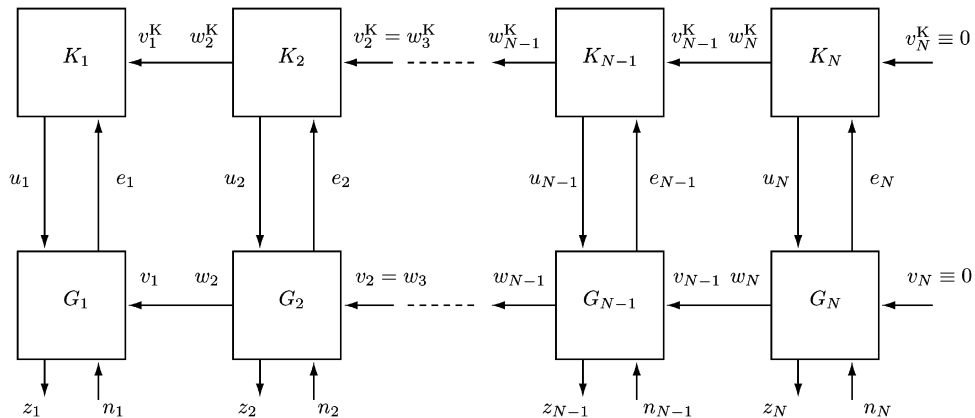


Fig. 18. Distributed control information structure.

V. CONCLUDING REMARKS

Only certain aspects of automating large-scale irrigation networks are considered in the preceding sections. In particular, attention is focussed on the following.

- 1) Channel modelling for control.
- 2) Water level control under:
 - a) decentralized feedback and feedforward control information structures;
 - b) a centralized control information structure;
 - c) a distributed generalized of the decentralized feedback control information structure.

For the decentralized schemes, a key design tradeoff is identified between:

- 1) local set-point regulation and disturbance rejection performance;
- 2) water-level error propagation performance.

An optimal control approach, based on the H_∞ loop-shaping generalization of classical ideas, is proposed to systematically deal with this tradeoff in the design of centralized and distributed compensators. Field tests confirm that our framework for modelling and controller design can yield excellent performance in terms of set-point regulation and load-disturbance rejection and, correspondingly, good service to farms while reducing water wastage.

The implementation of fully automated irrigation networks requires various other issues to be addressed. As mentioned in Section I, this includes supervisory network control to determine water-level set points on the basis of demand and network capacity, and other more low-level control issues such as anti-windup and bumpless transfer, performance monitoring, and fault-detection and reconfiguration, as briefly discussed below.

The regulating gates along a channel are subject to saturation. The gate positions range from fully open to fully closed. Simple anti-windup schemes [17] have been successfully employed for the field tests of the decentralized controllers designed as described. More advanced schemes have also been considered for centralized schemes in [39] and [40] and distributed schemes in [35]. These schemes also exhibit desirable bumpless transfer properties, which is important from the perspective of startup and shutdown.

Due to the large number of sensors and actuators in large-scale irrigation networks, faults are bound to happen from time to time. It is, therefore, important to have a mechanism to detect sensor and actuator fault events and to have procedures in place to minimize the consequences in some sense. Strategies to minimize the effect of a fault using reconfigured signals and controllers are described in [41]. It is shown that under natural conditions, suitably modified control objectives can be achieved in the presence of faults.

Finally, in a large-scale implementation of a decentralized control structure, at any given time it is possible for there to be a number of control loops which are (or have become) poorly tuned. This is particularly relevant in that the bad effects of poor local performance can propagate upstream. Correspondingly, a key issue is performance monitoring for the detection of poorly tuned control loops, so that these may be retuned. A corresponding approach, based on reference models of acceptable performance, is proposed in [42]. The method takes into account the interactions between control loops, and it has been successfully tested on the Haughton Main Channel. ■

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